Digital Calibration Certificates at Inmetro

Evolution 2024–2025: From Single-Point to Complex Tables

Gean Marcos Geronymo

Instituto Nacional de Metrologia, Qualidade e Tecnologia (Inmetro)
Duque de Caxias, Brazil
gmgeronymo@inmetro.gov.br

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Status 2024: Initial Framework

DCC Tools - First Version:

- **Scope:** Single laboratory (LAMPE)
- **Data Model:** Single measurement points only
- **Architecture:** AWS Lambda serverless
- **Output:** DCC XML with base64-embedded PDF
- **Use Case:** Proof-of-concept

Limitations:

- Could not handle complex calibration tables
- Single laboratory isolated implementation
- No production deployment

2025 Evolution: Framework Improvements

From Simple to Complex Data Structures:

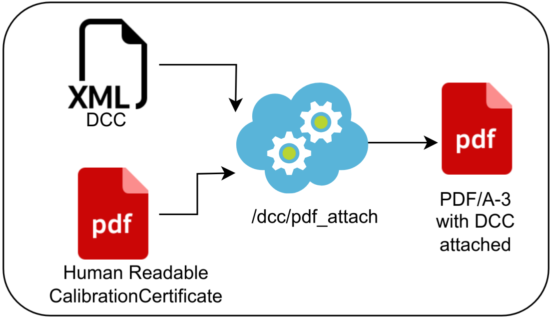
2024 (Before)	2025 (After)
Single measurement point	Multiple measurement points
Single quantity	Multiple quantities
No index support	Multiple indices (voltage, frequency, etc.)
Simple certificates	Complex calibration tables

Example: AC-DC transfer with indices for range, voltage, frequency

Architecture Evolution

Technical Improvements:

- **New Architecture:** Flask-based microservice with Docker
- **Input Options:** JSON (automation) + Excel (operators)
- **PDF/A-3 Approach:** XML embedded as attachment (METAS-based)
- **Endpoints:** Generate, validate, visualize, PDF attach



From Single Lab to Two-Laboratory Pilot

2024: Single Laboratory (LAMPE – Electrical Standards)

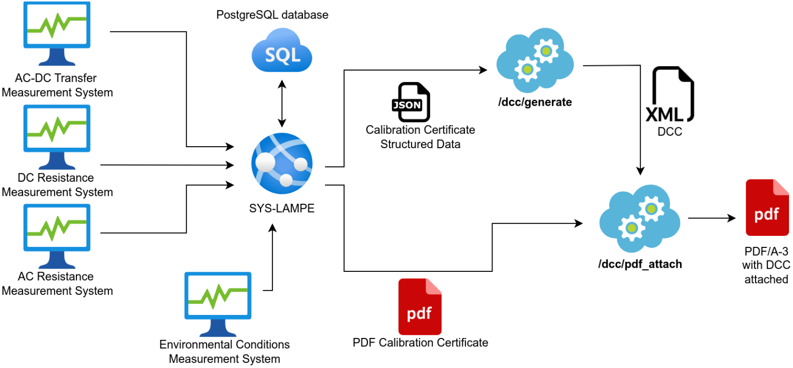
2025: Two-Laboratory Pilot

- **LAMPE:** Primary standards (provides traceability)
- **LACEL:** Calibration laboratory (receives traceability)

Integration Workflow:

1. LAMPE issues DCC to LACEL
2. LACEL automatically extracts calibration data
3. Updates reference standard databases
4. Uses data for calibrating customer equipment

End-to-End Integration



- Automated DCC generation from database
- PDF/A-3 with embedded DCC
- LabVIEW DCC reader for downstream processing
- Complete traceability chain

Future Perspective: 2026 Roadmap

Next Steps:

- **Q1-Q2 2026:** Deploy to external customers
- **Production:** Real DCCs issued to clients
- **Benefits:**
  - Eliminate manual transcription (up to 500 calibration points!)
  - Zero transcription errors
  - Real-time instrument validation

Harmonization Challenges:

- RefTypes vocabulary for electrical metrology
- Inmetro developing custom ontology
- International convergence via FORUM-MD-TG-H-DCC/DRMC

Summary: 2024 vs 2025 Evolution

Aspect	2024	2025
Laboratories	1 (LAMPE)	2 (LAMPE + LACEL)
Data Complexity	Single-point	Multi-quantity tables
Architecture	AWS Lambda	Flask + Docker
Deployment	Proof-of-concept	Production pilot
Next Step	Internal testing	External customers

Repository: `github.com/gmgeronymo/dcc_tools`

Thank you!

Questions?

Contact: gmgeronymo@inmetro.gov.br

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